(Breaking) Intergenerational Transmission of Mental Health

Bütikofer, Ginja, Karbownik, and Landaud (JHR 2023)

University of Georgia

April 5, 2026

Outline

The Global Mental Health Burden

- Mental health disorders affect **more than one billion people worldwide**
- Among leading causes of disability and rising share of global disease burden
- Economic costs are large and growing:
  - US: $201 billion in 2013; projected $225 billion by 2019
  - Norway: NOK32 billion ($3.7B USD) in 2013; NOK65 billion by 2023
- COVID-19 has further exacerbated demand for mental health care
- True economic costs exceed medical spending: productivity loss, forgone taxes, externalities on others

An Understudied Externality: Intergenerational Transmission

- One key externality is the intergenerational association between parent and child mental health
- If mental health conditions transmit across generations, costs imposed on future generations are substantial and unaccounted for
- Three research questions:
  - What is the **association** between parent and child mental health?
  - How different is it from physical health? How important is the **extended family**?
  - Can a **targeted low-touch policy** break the intergenerational link?
- Answers require: population-level data, objective health measures, family linkages across generations

Preview of Main Findings

- **Persistence is large:** having a parent with a mental health diagnosis raises child's probability of mental health diagnosis by 9.3 percentage points (40% of control mean)
- **Dynasty matters:** focusing only on parents understates persistence by 42% — extended family adds substantial explanatory power
- **Mental $\approx$ physical:** intergenerational associations in mental and physical health are broadly similar in magnitude; genetics likely not the full story
- **Policy can help:** a 2007 Norwegian pilot program reduced the parent-child mental health association by 39%
    - Larger effects for children of college-educated parents; raises equity concerns

Outline

Prior Literature on Intergenerational Health

- **General health persistence:** Andersen (2021), Björkegren et al. (2022), Halliday et al. (2021) — registry-based estimates tend to be smaller than survey-based
- **Specific conditions:** birth weight (Currie & Moretti 2003), cardiovascular (Lloyd-Jones et al. 2004), BMI (Classen 2010)
- **Mental health specifically is scarcer:**
  - Survey-based: Johnston, Schurer, & Shields (2013); Bencsik, Halliday, & Mazumder (2021); Vera-Toscano & Brown (2021)
  - Specific disorders: Akbulut-Yuksel & Kugler (2016) for depression; Eley et al. (2015) for anxiety
  - Almost all rely on **self-reported survey data**: recall bias, small samples, ordered intervals
- Survey estimates of parent-child mental health ICs: 0.13 (Johnston et al.) to 0.22 (Bencsik et al.)
- Only Andersen (2021) provides population-level estimates using administrative GP/hospitalization data for Denmark

This Paper's Contribution

- **Administrative, population-level data** for Norway: near-universal coverage, no recall bias, no small-sample issues
- **Extended family (dynastic) analysis:** following Adermon, Lindahl, and Palme (2021) for education, but extending to health — aunts, uncles, cousins, and spouses thereof
    - Ignoring dynasty **understates** intergenerational persistence by 42%
- **Both mental and physical health** measured with the same data infrastructure
- **Causal policy evaluation:** triple differences design leveraging a targeted Norwegian pilot program
- Provides a lower bound for intergenerational persistence (sickness leave truncated from top and bottom — see Data section)

This Paper's Contribution (cont.)

- Connects three literatures:
  1. **Intergenerational health persistence**: adds administrative mental health, dynasty component, causal policy variation
  2. **Dynastic human capital** (Adermon, Lindahl & Palme 2021): replicates for Norway, extends beyond education to health outcomes
  3. **Health interventions in childhood**: Hjørt, Sølvsten, & Wüst (2017); Bütikofer & Salvanes (2020); Baranov et al. (2020) — light-touch policies can reduce persistent disadvantage
- Broader implication: standard program evaluations that stop at the directly treated generation **may understate returns** to mental health investments

Outline

Why Norway?

- Healthcare is **highly subsidized and universally accessible** — limits selection into treatment based on income
- Unique **personal and family identifiers** across all administrative registers: link individuals across four generations
- General practitioners (GPs) and primary care ER doctors are **obligated to report all consultations**
  - Near-universal coverage for children 13–18: 91% visited primary healthcare before seeing a specialist
- Reduces concerns about underreporting relative to survey data:
  - Bharadwaj, Pai, and Suzie-delyte (2017): large underreporting of MH in surveys vs. administrative records
- Private insurance/clinics have negligible market share ($\approx$ 10,000 adults with private coverage in 2003)

Data Sources

- **Family registers (1992–2015):** link parents, children, and extended family via birth certificates; construct extended family horizontally (siblings, cousins, spouses)
- **KUHR — Control and Payment of Health Refunds (2006–2020):** children's health
  - GP and primary care ER visits; ICPC-2 coded
  - Available from 2006 onward
- **Social Security sickness leave registry (1992+):** parental health
  - All absences > 3 days certified by a physician; ICPC-2 coded
- **Demographic, education, and income registers:** controls and heterogeneity analyses

Measuring Children's Mental Health

- **Source:** KUHR registry — GP and ER visits between ages 13 and 18
- **Mental health indicator:** any GP or ER visit with an ICPC-2 code starting with the letter "P"
- **Extensive margin:** indicator for at least one mental health event
    - Control group mean: 23% of adolescents whose parents have no MH diagnosis have at least one MH event
    - 24% of all children in the sample have a MH diagnosis
- **Intensive margin:** total number of MH-related healthcare events
- KUHR not available before 2006 $\Rightarrow$ children observed ages 13–18 between 2006–2020; born 1988–2007
- Nearly all children (94.3%) have at least one GP visit between ages 13–18 — near-universal coverage

Measuring Parental Mental Health: The Sickness Leave Registry

- **What it captures:** any sickness leave certified by a physician for mental health reasons (ICPC-2 code "P"), measured when parents are ages 25–30
  - Parent ages 25–30 are observed prior to child's adolescence, ruling out co-diagnosis
- **Extensive margin:** indicator for at least one MH sickness leave event
- **Intensive margin:** average number of MH sickness leave days
- **Why this window:** ensures parental health is measured *before* the child's outcome window (ages 13–18)
- 7% of mothers and 4% of fathers take at least one MH sickness leave at ages 25–30
- Most common MH condition: depression (both genders)

Limitations of the Sickness Leave Measure

- **Eligibility constraint:** only individuals in the labor market qualify — 6% of 25–30 year olds ineligible
  - Ineligibility more common among foreign-born, lower-income individuals
  - Our estimates likely represent an **upper bound** for severe conditions (sickest may not work); a **lower bound** for mild conditions (mild cases may not take leave)
- **Severity truncation:** not all mental health episodes lead to leave — depression manageable by medication may not appear
- **Coverage overlap:** for 2006–2008, compare sickness leave to KUHR data: 23% of individuals with a MH event in KUHR took sickness leave
- **External validity:** parents with MH diagnoses are somewhat more likely to be parents than those without, suggesting our sample is not severely selected
- Robustness: control for sickness leave eligibility determinants (education, income, immigration status) — results unchanged

Constructing the Extended Family (Dynasty)

- Following Adermon, Lindahl, and Palme (2021): define the dynasty to include **five branches** of the extended family beyond parents:
  1. **Parents' siblings** (sp) — 25% genes shared
  2. **Spouses of parents' siblings** (ssp) — plausibly unrelated genetically
  3. **Siblings of spouses of parents' siblings** (sssp) — plausibly unrelated
  4. **Parents' cousins** (cp) — 12.5% genes shared
  5. **Spouses of parents' cousins** (scp) — plausibly unrelated
- Associations for "plausibly unrelated" members (ssp, sssp, scp) constitute 18–15% of total intergenerational persistence $\Rightarrow$ genetics alone cannot explain the full picture
- Control for **number of members** in each branch of extended family to avoid mechanical effects

Sample Selection and Descriptive Statistics

- Start: all children born in Norway 1988–2007, resident ages 13–18 in 2006–2020 ($N = 732{,}437$)
- Restrict to:
  - Parents with labor income at ages 25–30 (sickness leave eligibility): drop 4,907
  - Children with known grandparents and great-grandparents: $N = 370{,}498$ preferred sample
- **Children:** 24% have any MH diagnosis; 6.96% have depression; 51% male
- **Mothers:** 54% take at least one sickness leave ages 25–30; 7% for MH reasons; most common reason: musculoskeletal (23%)
- **Fathers:** 28% take at least one sickness leave; 4% for MH reasons
- Parent-child MH correlations below 0.04 in the raw data (Pearson) — associations are real but moderate

Outline

Econometric Model: Intergenerational Persistence

Regress child mental health on parental and extended family mental health:

$$Y_{it} = \alpha + \beta_p Y^{p}_{t-1} + \sum_{k \in \{sp,\, cp,\, ssp,\, sssp,\, scp\}} \beta_k Y^{k}_{t-1} + \gamma \mathbf{X}_{it} + \varepsilon_{it}$$

- Y_{it}: indicator (extensive) or count (intensive) of child's MH events, ages 13–18, $\times 100$
- Y^{p}_{t-1}: parental MH sickness leave, ages 25–30 (measured before child outcome)
- Y^{k}_{t-1}: MH sickness leave of extended family branch k
- $\mathbf{X}_{it}$: year of birth FE, year first observed in KUHR, gender, number of individuals per dynasty branch; *select specs* add SES controls
- Parameters of interest: β_p (parent-child) and β_k (extended family)
- Standard errors: Eicker-Huber-White, clustered at the family level

Identification: What We Can and Cannot Claim

- **What the design delivers:** associations / correlations between parent and child MH, not causal effects of parental MH on child MH
- These associations are policy-relevant regardless of mechanism:
  - Mental health diagnoses trigger initial treatment and follow-up costs
  - They motivate the targeted intervention in Section V
- **Timing:** parental health measured at ages 25–30, strictly before child's outcome window (ages 13–18) $\Rightarrow$ rules out co-diagnosis
- **Not causal but informative:** associations are robust to controlling for SES, school FE, GP FE, prenatal health, and birth characteristics
- Cannot separate genetic from social/environmental channels — dynasty analysis provides suggestive evidence that social factors matter

Robustness Strategy

- Eight robustness columns (Tables 7–8), from most to least parsimonious:
  1. Baseline (Column 6 of main tables)
  2. No controls
  3. Add primary care visit frequency (GP visit FE) — addresses surveillance bias
  4. Add sickness leave eligibility determinants and fertility controls
  5. Add individual-level controls: birth weight, birth rank, parent ages at birth
  6. Add school-by-cohort FE
  7. Add first GP fixed effects (at age 13)
  8. Extend parental age window to 30–35
- Results: parent-child association declines by up to 40% with controls, but remains statistically significant and economically meaningful throughout
- Largest declines from SES controls (3.4 pp) and GP visit controls (1.9 pp)

A Note on Measurement of Parental Health

- **Upper vs. lower bound concern:**
  - If milder conditions have lower persistence $\Rightarrow$ sicker individuals more likely captured $\Rightarrow$ estimates are an **upper bound**
  - If sickest individuals exit the labor market $\Rightarrow$ most severe cases missed $\Rightarrow$ estimates are a **lower bound**
- Authors conclude: upper bound concern is more plausible given the data
- **Validate** with 2006–2008 overlap of KUHR and sickness leave: 23% of individuals with a MH symptom in KUHR took sickness leave
- **Administrative vs. survey:** Bharadwaj et al. (2017) show large underreporting of MH in surveys — administrative records likely more accurate

Outline

Main Results: Extensive Margin (Mental Health)

- **Parent-child association (Column 1):** 9.8 percentage points
  - Control group mean = 23% ⇒ this is a **40% increase** in the probability of a MH event
- Adding extended family branches (Columns 2–5): parent coefficient barely changes
  - Parent-child association is largely **orthogonal** to extended family associations
- **Full dynastic model (Column 6):** parent coefficient = 9.6 pp; sum of all extended family = 7.6 pp ⇒ dynasty adds 79% more
- **Controlling for physical health (Column 7):** parent coefficient = 9.3 pp — MH associations not driven by cross-condition correlations
- All extended family branches are statistically significant; magnitude decreasing in familial distance

Main Results: Intensive Margin

- Substitute indicator variables with **standardized counts** of MH health events (children) and MH sickness leave days (parents and extended family)
- **Parent-child association:** 0.05 standard deviations (extensive: ≈ 0.05 too) — broadly comparable
- **Extended family:** sum of coefficients $= 0.037$, or 75.5% of the parent-child association
  - Very similar ratio to the extensive margin
- Spouses of parents' cousins (scp) becomes statistically insignificant in the intensive margin saturated model
- Physical health intensive margin: sum of extended family $= 76.3\%$ of parent-child association — nearly identical to mental health
- Intergenerational health elasticities (intensive margin): **0.05 for MH**, 0.06 for physical health
  - Grows to 0.09–0.10 when dynasty is included
  - Substantially smaller than socioeconomic outcomes (income 0.34, wealth 0.35, education 0.25)

Dynastic Effects: Why the Extended Family Matters

- Ignoring dynasty **underestimates** total intergenerational persistence by 42%
- Extended family associations decrease with genetic distance:
    - Parents' siblings (sp): 14.8% of parent-child association
    - Spouses of parents' siblings (ssp): 7.1% — plausibly unrelated genetically
    - Spouses of parents' siblings and their siblings: smallest, but still significant
- "Plausibly unrelated" members (ssp, sssp, scp) account for 18% of total extensive MH persistence
    - Suggests social mechanisms (modeling behavior, resources, awareness) matter — not purely genetic
- Contrast with **educational persistence:** dynastic educational associations constitute only 6% of the corresponding parent-child association
- Mental health is more "visible" within extended families than education — may enable social transmission

Physical vs. Mental Health Persistence

- **Prior literature** often assumes mental $\neq$ physical health in intergenerational persistence (different mechanisms, heritability)
- **This paper's finding:** MH and physical health (OH) associations are largely similar
  - Extensive margin: parent-child MH IC $= 0.050$, OH IC $= 0.062$ — modestly larger for physical health
  - Intensive margin: nearly identical magnitudes
- MH associations are unaffected by controlling for physical health of generation $t-1$ (and vice versa)
- Extended family patterns mirror each other: parent-child association independent of dynasty for both MH and OH
- Implication: the mechanisms driving intergenerational health persistence — genetic, social, environmental — are not fundamentally different across MH and OH

Comparing Health and Educational Persistence (Table 6)

- Replicate Adermon, Lindahl, and Palme (2021) for Norway using Grade 10 GPA as outcome
- **Parent-child education elasticity:** 0.45 — nine times larger than MH (0.05)
- **Dynasty sum for education:** 0.549 in Norway (vs. 0.518 in Sweden)
- Dynastic education associations **do not change** even when controlling for dynastic health associations
  - Education and health channels are separate — health does not mediate educational transmission
- "Genetically unrelated" dynastic members matter more for health (18%) than for education (6%)
  - Mental and physical health issues may be more "visible" within extended families than education, enabling social transmission

Magnitudes Relative to Prior Literature

- Administrative data estimates tend to be **two to four times smaller** than survey estimates
  - U.K. survey ICs: 0.13–0.22 (Johnston et al.; Bencsik et al.)
  - This paper (Norway, admin): parent-child IC ≈ 0.05 (intensive margin)
- Consistent with other registry-based work:
  - Björkegren et al. (2022): 0.11–0.14 using hospitalizations in Sweden
  - Andersen (2021): 0.09–0.14 for general health in Denmark
- Likely explanations for gap vs. surveys:
  - Upward bias from recall, subjective perception, Likert scale in survey data
  - Norway's universal healthcare reduces selection into treatment
- Norway is not an outlier: OECD data suggest comparable MH burden to other developed countries

Outline

The 2007 Pilot Program: Modellkommuneforsøket

- In 2007, the Norwegian Directorate for Children, Youth, and Family Affairs launched a **pilot in 26 municipalities**
- Goal: find best practices for supporting young children (ages 0–6) of parents with mental health conditions
- Municipalities received NOK100,000–290,000 annually from central government
- The program was **light-touch and varied** across municipalities:
  - Screening tools for psychological distress among targeted children
  - Establishing specialist teams and interagency cooperation
  - Childcare center psychologists; home visits; parenting programs (PMTO)
  - Family coordination and case managers
- By 2010, all pilot municipalities had a registration system and screening tools
- Success led to a **2010 national law** mandating similar practices across all Norwegian municipalities

Who Was Targeted and Why It Matters

- **Direct target:** children ages 0–6 in 2007 whose parents had a MH diagnosis or substance abuse problem
- Healthcare providers were mandated to: register dependent children, converse with patients about children's needs, offer help navigating support services
- **Not a screening program for children themselves:** children were not required to be screened for MH conditions
  - This creates an ambiguous prediction: more scrutiny could increase diagnoses; effective treatment could decrease them
- Empirically: the policy **reduced** the parent-child association $\Rightarrow$ treatment effect dominates the increased scrutiny effect
- Qualitatively evaluated as successful in 2010 and 2014 (Deloitte 2015; Rambøll 2010)

Empirical Strategy: Triple Differences

Since 26 out of 428 municipalities were not chosen randomly, use a **triple differences** design:

$$Y_{icm} = \alpha+\beta(E_{ic}\times D_{im}\times MPH_i)+\delta_1 MPH_i+\delta_2(E_{ic}\times MPH_i)+\delta_3(D_{im}\times MPH_i)+\delta_4(E_{ic}\times D_{im})+\mathbf{X}_i'\boldsymbol{\mu}+\rho_c+\pi_m+\varepsilon$$

- Y_{icm}: any MH GP/ER visit ages 13–18 for child i, cohort c, municipality m ($\times$100)
- MPH_i: indicator for parental MH diagnosis (2000–2010)
- D_{im}: indicator for treated municipality
- E_{ic}: indicator for child aged ≤ 6 in 2007 (eligible cohort)
- β: ITT estimate of how the program affects the parent-child MH association
- Controls: year of birth FE, municipality FE, child gender, parent ages, birth weight, birth rank, parental education
- Municipalities matched via Mahalanobis distance on pre-intervention characteristics

Credibility: Parallel Trends and Placebo Tests

- **Identifying assumption:** absent the pilot, children with and without MH parents would have evolved similarly in treated and control municipalities
- **Event study (Equation 4):** estimate β^τ for each birth cohort age in 2007
  - No systematic pre-trends for ages 7–11 in 2007 $\Rightarrow$ supports parallel trends
  - Largest treatment effects for youngest children (ages 2–4): longest exposure to program
- **Placebo 1:** outcomes predetermined at birth (e.g., birth weight) — triple interaction is never statistically significant
- **Placebo 2:** outcomes unlikely to be affected by intervention (fractures, musculoskeletal) — triple interaction statistically insignificant
- **Robustness:** add municipality-specific trends in supply of health services; results unchanged (Panel C of Table 9)

Policy Effects on the Intergenerational Association

- **Baseline parent-child association** (control municipalities, ineligible cohort): 9.7 pp
  - Very close to 9.8 pp in the full-population analysis
- **Treatment effect ($\hat{\beta}$):** -3.8 pp (statistically significant at 1%)
  - Program **reduced the parent-child mental health association by 39%**
- Robust to intensive margin transformation (Panel B): -1.1 sickness leave days per year
- Robust to controlling for municipal trends in health services (Panel C)
- Most affected: youngest children at the start of the program (ages 2–4 in 2007)
  - Longer exposure and earlier age $\Rightarrow$ larger effects

Heterogeneity: Who Benefits Most?

- **By parental education:**
  - College-educated: -6.2 pp treatment effect (significant at 1%)
  - Non-college: -2.1 pp (statistically insignificant)
- Pilot was more effective for children of college-educated parents
  - Possibly: better ability to navigate and utilize offered services
  - Policy implication: **reduces average** parent-child MH association but **increases within-generation inequality** in mental health outcomes
- **By child gender:** broadly similar effects for boys and girls on extensive margin; 77% larger for boys on intensive margin
  - Boys may be more responsive to positive inputs, especially in childhood (Autor et al. 2019)
- Authors flag this as a **cautionary tale** for policymakers: interventions that reduce average persistence can still widen inequality within the affected generation

Outline

Main Takeaways

- **Persistence is substantial:** parental MH diagnosis raises child MH probability by 9.3 pp (40% of control mean); stable across extensive and intensive margins and to extensive robustness checks
- **Dynasty matters:** focusing only on parents understates intergenerational mental health persistence by 42%; social (not just genetic) transmission is implicated
- **Mental $\approx$ physical:** MH and physical health intergenerational associations are broadly similar — mental health is not uniquely persistent or uniquely genetic
- **Policy can work:** a low-touch Norwegian pilot reduced the parent-child MH association by 39%, with largest effects for youngest children
- **But equity matters:** gains accrued primarily to children of college-educated parents; the pilot may have increased inequality within the treated generation

Policy Implications and Future Directions

- **Implication 1:** Standard program evaluations that stop at the directly treated generation understate the returns to mental health investments
- **Implication 2:** Light-touch, early-childhood interventions targeting children of MH-diagnosed parents can be cost-effective — the 2010 national law scaled this approach
- **Implication 3:** Accounting for extended family links is important for understanding the *full* burden of mental illness on society
- **Open questions for future research:**
  - Causal mechanisms: exogenous shocks to parental MH (e.g., job loss, health shocks) on child outcomes
  - Long-run effects of the pilot on education, income, and adult MH
  - Do the gains from the program extend beyond mental health to labor market outcomes?
  - Optimal design of interventions that reduce average persistence *and* inequality simultaneously